

User-centred design of Web sites and the redesign of LineOne

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This paper describes the results of a study of the LineOne Internet site by members of the User Centred Design group. As a result of the study the site was redesigned. It also gives examples of successful Internet sites and looks at the components which contribute to this success. The original LineOne home page was seen as cluttered and confusing by Focus group participants. There were too many menus and women commented that, with news and sport dominating the home page, it seemed to be aimed at men. Many wanted to customise, for example, the news, so that items of interest to them were highlighted. The right hand menu was not visible on standard 640 × 480 pixel screens, which was one of the reasons why people were not accessing many areas of the site. Some of the titles on the menu bar were also fairly obscure. The ability to search the site from the top level was seen as desirable by those interviewed. As a result of the work done by the User Centred Design group, the home page was redesigned. Users can now set filters to further customise the information they receive, such as news and entertainment, and the site has received very favourable reviews in the specialist press.

1. Introduction

The LineOne Internet service was launched in April 1997. It is a value-add service aimed at the leisure market and includes areas on current news, entertainment information, share prices and money matters as well as games and areas for children and teenagers. Within a month of launch it was clear that users were not accessing many of the areas within the site and that a radical redesign would be needed. The User Centred Design group was asked to examine the site and come up with recommendations for improvements. You can seldom anticipate what members of the public will say and the comments gathered during the Focus group interviews were both entertaining and informative.

A total of nine Focus groups were run over a period of two weeks in June 1997. The people interviewed had a variety of Internet and computer experience. Some were mothers who had used a personal computer only a few times and had never used the Internet. Others were experienced Internet users and some were existing LineOne users.

During the course of our work we were very much aware that Internet page design is still a relatively new art and it is often difficult to define what gives a Web site that extra something that keeps people coming back for more. Some, like OneWorld, contain few graphics, but are well laid out and full of useful information for the target audience; others, like the Pepperami site, are just good fun (see Figs 1 and 2).

2. Designing for the user

Before looking at the detail of the LineOne review there are some general principles which need to be borne in mind. A number of aspects need to be considered when designing a Web site.

- Who is the user?

Age and gender and whether the site is being used for leisure or business will all have an effect on the way the site is designed and laid out, as will the user's experience, or inexperience, of the Internet,

- Usefulness of site content

With the above in mind how useful will the user find the site?

- Design

How should the site be designed to appeal to the target audience? Does the site conform to the company brand image?

- Usability

How easy is it to get around the site? Are menu titles meaningful? Is the registration process easy to follow?



Fig 1 OneWorld Web site.

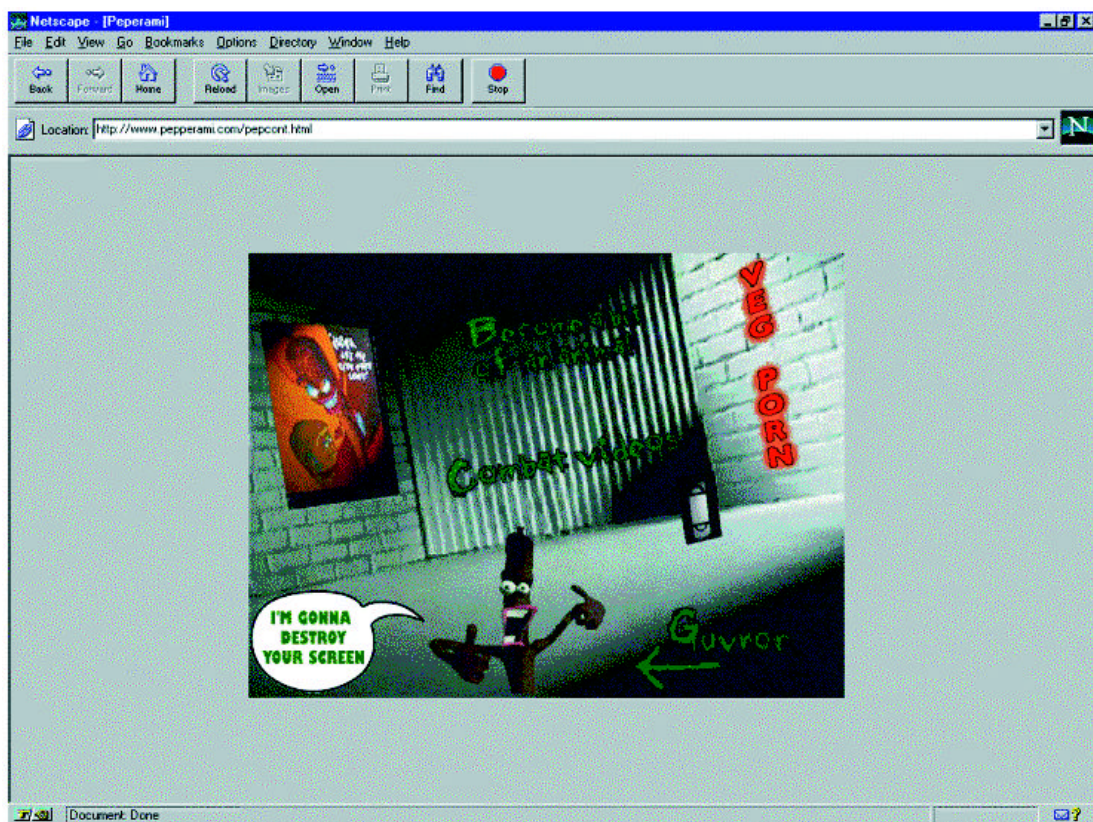


Fig 2 Pepperami Web site.

Figure 3 illustrates these interdependent variables. As the user changes, the other three aspects may also need to change. However, for a site like LineOne, there is a need to appeal to a number of different audiences at the same time.

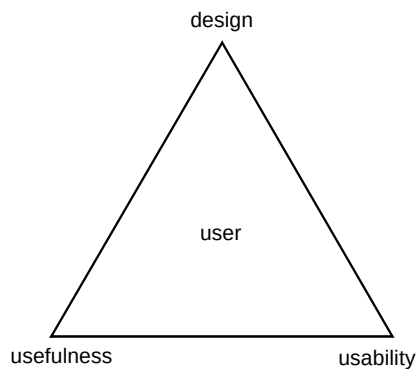


Fig 3 Relationship between variables.

Therefore, whether the user seeks out a site, and then subsequently returns to it, will be dependent on its design, usefulness and usability. In some cases the usefulness of a site may override the importance of the other two aspects. In a competitive environment, however, services like LineOne need to be at least as good, and preferably better, on these three essentials of good Web design if they are to attract and retain customers.

3. Focus group results

The Focus groups explored a number of issues including:

- registration,
- home page design,
- navigation,
- content,
- alternative/competitors' home page designs.

The results are described below.

3.1 Registration

A free CD had been given away with the Sunday Times and contained information about the LineOne service with the opportunity to register. People had been signing up for the free month's trial and then failing to register. A number of problems with the registration process were discovered, including unintelligible error messages. Because many of the potential users are novices as regards the Internet, it is essential that they are guided through registration by friendly error messages which tell them what to do next. The CD promised to be jargon-free, but it was not. In one section the CD talked about modem speeds needed. One interviewee did not know what a modem was and we would

expect this to be a fairly typical response from inexperienced computer users.

Several people would have liked a direct telephone line service, where they rang up and someone else filled in the form for them. They were also concerned about giving their credit card details over the Internet and needed reassurance that the connection was secure. They would have been happier giving details over the telephone.

3.2 Home page design

Paper copies of the home page were given to each member of the Focus group (see Fig 4). They were asked for their comments under the following headings:

- overall design of the home page,
- target audience,
- intelligibility of icons or links,
- suggestions for improvements.

Overall design

Some of the current LineOne users had 640 × 480 pixel screens and could not see the menu on the right hand side on their own computers. This was an obvious clue as to why people had not been accessing many of the areas. Another comment was that there was far too much clutter on the home page and it was difficult to see what was on there. The content seemed very useful, but was hard to find.

Target audience

Because *The Sun* newspaper was branded so prominently on the home page one person said it looked like *The Sun* home page and that it would appeal to 'The man on the tube reading *The Sun* and drinking a beer'. News and sport also seemed to dominate the content. As a result the site was seen both as too downmarket and as not appealing to women.

Intelligibility of icons or links

The headings on the right hand menu were studied and interviewees were asked what they thought each heading meant. Here are some of the suggestions.

- Connect — 'Is this a dating agency?' (actually Internet access),
- Out of this World — 'Is this about outer space or the X files?' (actually astrology),
- 1015 — 'Isn't this the specific gravity for a beer?' or 'Is this the 10.15 watershed?' (actually an area for 10—15 year olds).



Fig 4 LineOne before the changes.

Some people thought that the mouse icon at the bottom of the page looked like a condom, which did not support the type of quality image that BT would wish to project!

Also people did not want to work hard at interpreting the meaning of link titles. They were asked for their suggestions for new titles, and these were implemented in the redesign.

Navigation

Once people had found the 'Home' button they felt fairly secure. One of the recommendations made was that the Home button should be made more prominent. The redesign was an opportunity to simplify the design of the pages and make navigation easier.

Content

Many people felt, by the end of the session, that the content was very useful, but was not always easy to find. They wanted to be able to customise some of the information received, and to be alerted, for example, to news on a particular subject. They also wanted a search facility, so that they did not have to spend a long time looking for specific areas.

There were several small-business people in the Focus groups who found the Money Minder section extremely useful.

Indication of time on-line

Many people said that they would like to know how much time they had spent on-line. This facility was already offered by competitors.

3.3 Alternative/competitors' home pages

Participants were asked to give examples of sites that they felt were well designed. The AOL site (see Fig 5) and redesigned CompuServe site (see Fig 6) both received favourable comments and were used as models when the home page was being redesigned.

Following the launch of Version 2 in autumn 1997 *.net* magazine included the following comments in its review of the service:

'Version 2 is bolder and better with a significantly improved interface and lots of dandy new features. Intelligent agents... have been made an integral part of the service to enable subscribers to customise LineOne.'

The home page (Fig 7) has a much cleaner look and the titles are much simpler. It no longer looks as if it was designed solely for men, but has general appeal to adults of both sexes, and to children. Time on-line is now displayed at the bottom right of the screen and there are more customisable areas within LineOne, as requested by customers.



Fig 5 AOL home page.



Fig 6 CompuServe home page.



Fig 7 LineOne after the redesign.

3.4 Cost benefits

The cost of calls to helplines can be enormous. With thousands of subscribers potentially calling the helpline the profits from offering the service could soon be eroded, even if each call only takes a few minutes of a customer service advisor's time.

It is far cheaper to find out possible user problems during the development phase; not to mention the greater customer satisfaction arising from the experience of using a well-designed service. It is vital that BT's Internet products are well designed if it is to retain its reputation as a quality company in a competitive market-place.

4. Further information about user-centred design

User-centred design is particularly appropriate to specific kinds of product and service. It makes use of a range of techniques to understand users' needs and capabilities, including Focus groups and user testing, and then translates these into a design solution. Typically, it follows an iterative process consisting of rapid needs elicitation, concept design and user testing.

4.1 What services benefit from user-centred design?

- Commodity products which are not differentiated from the competition by cost, features or reliability can be given competitive advantage if they are easy to use. An example of this from BT's portfolio is the Freestyle 1200 cordless phone.
- Services which may generate large numbers of calls to helplines, or a high degree of customer churn owing to dissatisfaction, will benefit from being designed with the users' needs paramount.

In particular, an analysis of what typical tasks the user wants to do can repay many benefits when it comes to laying out a Web site or designing a user interface to a Windows 95 application. A good example of this is the LineOne service. This experience has been distilled into a set of guidelines 'Designing Web Sites, Guidelines for Usability', which are available at: <http://www.pase.bt.co.uk/ucdg/uiguide.html>

- Services whose use is time critical and/or which demand a high degree of error-free performance need to be carefully crafted to take into account the user's experience and the tasks they are trying to undertake, e.g. the SMART service used by BT's 150 customer service advisors.



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